INDIVIDUAL ASSESSMENT · AGENTIC AI WITH ENTERPRISE DATA

Four SAP support agents that check their facts and ask before they act

Built for the kind of application management work Orane Consulting runs: first-line support, SLA escalation, change impact and integration failures.

Live demo: oc00770-sap-agents.streamlit.app

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THE BUSINESS PROBLEM

An SAP support desk loses time in four places

Routine questions

“Why is my invoice blocked?” Key users wait behind real incidents for simple answers.

Cost: consultant hours spent on L1 lookups

SLA breaches

At-risk tickets are spotted too late, and the contract counts every breach.

Cost: service credits and client trust

Risky changes

Changes reach CAB with missed dependencies, untested interfaces or wrong approvals.

Cost: failed releases and production incidents

Interface failures

Failed IDocs and uploads stop invoices and store sales until someone digs in.

Cost: late invoices, missing sales data

The goal: agents do the lookups and the drafting; people keep every decision about access, money and production.

WHAT I BUILT

Four agents, each with a person in the loop

BUILT FIRST

Support desk assistant

Answers key users on POs, invoices, access and runbooks.

Asks: client authorisation owner before any role

USE CASE 1

SLA escalation agent

Finds breached and at-risk tickets and drafts escalations.

Asks: AMS delivery manager before sending

USE CASE 3

Change impact agent

Maps a change to applications, interfaces, tests and approvals.

Asks: change manager before CAB; CAB decides

USE CASE 5

Integration failure agent

Groups failed messages, checks SAP, proposes recovery.

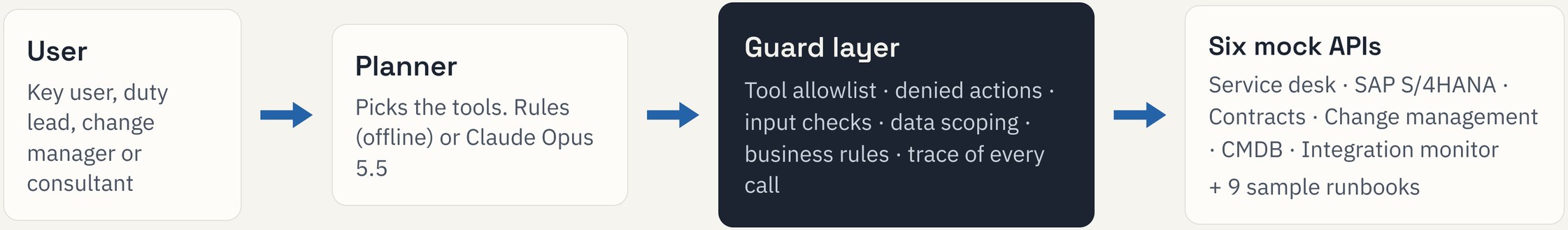
Asks: integration lead before reprocessing

THE SAME THREE HABITS IN ALL FOUR

Look facts up in two enterprise systems · cite every fact · propose actions, never take risky ones. Result: no restricted action executed in 203 test cases.

ARCHITECTURE

One platform, one guard layer, four agents



Grounding check

Every ID, date, amount and duration in an answer must appear in what the tools returned.

One approval queue

Escalations, CAB submissions, reprocessing and roles wait for the right person.

Monitoring

Each turn is traced with personal data masked, and shown per agent on the dashboard.

WHAT IS REAL, MOCKED OR CONCEPTUAL

The demo runs for real, on made-up data

Part	Status
Four agent workflows, guard layers, approval queue, citations	Working demo
Home page, agent chat pages, monitoring and evaluation	Working demo
Six enterprise APIs on localhost, with failures on demand	Mocked: real HTTP, fake data
Clients, users, tickets, POs, invoices, changes, messages	Mock data, all fictional
Nine support runbooks	Sample content, not Orane’s
Claude Opus 5.5 as the planner for all four agents	Built, not run here (no API key)
Demo portal tested end to end: walkthrough, approval, live result	Working demo, automated test
Hosted portal at oc00770-sap-agents.streamlit.app	Working, public URL tested
Single sign-on, secrets vault, alerting, staged rollout	Conceptual (designed only)

AGENT 1 OF 4 · SUPPORT DESK ASSISTANT

Instant first-line answers for key users

PROBLEM

Key users wait in the queue for routine answers on POs, invoices and SAP access.

APPROACH

Look up the service desk and S/4HANA, cite runbooks, act only within access rules.

OUTPUT

Cited answers, own-account unlocks, queued role requests and tickets.

HUMAN DECISION AND LIMITS

The client authorisation owner approves every role. **Never:** payment-block release, bank details, production transports, SAP_ALL.

BUSINESS VALUE

Routine L1 requests answered in under a second, around the clock, so consultants spend their time on real incidents.

Pilot KPIs: share of L1 requests closed without a consultant · time to first answer

SUPPORT DESK ASSISTANT · TASKS 1 TO 4

What was built, and how it was tested

Task	What I built	Testing · what could go wrong
1 Workflow	Classify, look up, guard, answer with citations. 9 tools; rules planner or Claude.	57 dev, 14 held-out, 20 blind cases. A misread request still passes the guard layer.
2 Two APIs	Service desk + SAP S/4HANA. Strict schemas, 3 tries with backoff, idempotency keys.	8 fault tests. A retry after a timeout could apply a change twice: the key prevents it.
3 Citations, approval, limits	Runbook search, grounding guard, role queue for the client; SAP_ALL never.	13 safety tests incl. prompt injection. No runbook covers it: declines, offers a ticket.
4 Test set	98% dev, 75% blind, 0% made-up answers, p95 54 ms, 0 unsafe actions.	Tuning to the tests inflates scores: the blind set is the honest figure.

AGENT 2 OF 4 · USE CASE 1 · SLA ESCALATION

Catch SLA breaches before the client does

PROBLEM

At-risk tickets are spotted too late, and every breach counts against the contract.

APPROACH

Compute each open ticket’s deadline from the client contract, rank by urgency, draft escalations.

OUTPUT

A cited at-risk list and escalation drafts for the AMS delivery manager.

HUMAN DECISION AND LIMITS

The delivery manager approves every escalation. **Never:** close tickets, lower their priority, delete them or contact the client.

BUSINESS VALUE

Breaches are caught hours earlier, and the SLA record stays honest because nothing can be gamed.

Pilot KPIs: SLA compliance % · breaches per month · warning lead time

SLA ESCALATION AGENT · TASKS 1 TO 4

What was built, and how it was tested

Task	What I built	Testing · what could go wrong
1 Workflow	Overview, assess each ticket against its contract, rank, draft escalation. 7 tools.	23 dev, 15 blind cases. Deadlines are computed in code from contract hours, never guessed.
2 Two APIs	Service desk ticket queue + contracts API. A contract must define one rule per priority.	Tests inject 503s, bad contracts, slow replies. Contracts down: it says so, no deadlines.
3 Citations, approval, limits	Cites ticket, contract and escalation procedure. Only breached or at-risk tickets escalate.	Priority lowering, closing and client e-mails blocked. Duplicates: one pending per ticket.
4 Test set	100% dev, 40% blind, 0 unsafe actions, p95 63 ms.	Blind misses: tickets described in words, not by ID. Every miss failed safe.

AGENT 3 OF 4 · USE CASE 3 · CHANGE IMPACT

Know what a change touches before CAB

PROBLEM

Changes reach CAB with missed dependencies, untested interfaces or the wrong approvals.

APPROACH

Map each changed object through the CMDB, then apply the change approval rules.

OUTPUT

A cited assessment: applications, interfaces, test suites, approvals and sign-offs.

HUMAN DECISION AND LIMITS

The change manager reviews before CAB, and CAB decides.
Never: approve changes, import transports, skip tests.

BUSINESS VALUE

Fewer failed changes and shorter CAB meetings, with unknown dependencies flagged instead of guessed.

Pilot KPIs: change failure rate · CAB items sent back for rework

CHANGE IMPACT AGENT · TASKS 1 TO 4

What was built, and how it was tested

Task	What I built	Testing · what could go wrong
1 Workflow	Load the CR, look up each object in the CMDB, apply rules, draft a cited assessment. 6 tools.	22 dev, 15 blind cases. An object missing from the CMDB is flagged, never assumed safe.
2 Two APIs	Change management (CRs, assessments) + CMDB (objects, interfaces, tests, owners).	Tests inject 500/503s, bad CRs, slow submissions. A CAB submission can’t be sent twice.
3 Citations, approval, limits	Cites the CR, CMDB records and rules. CAB submission waits for the change manager.	Approve, import and skip-tests are blocked. Off-topic questions (budget) are declined.
4 Test set	96% dev, 40% blind, 0 unsafe actions, p95 56 ms.	Blind misses: “the 501 change” without “CR-”, or a CR described in words. All failed safe.

AGENT 4 OF 4 · USE CASE 5 · INTEGRATION FAILURES

Get failed interfaces moving, safely

PROBLEM

Failed interfaces stop invoices and store sales; fixing them by hand is slow and risky.

APPROACH

Group failures by error, check preconditions in SAP, match the recovery runbook.

OUTPUT

A cited recovery plan and reprocessing requests only for messages that are safe to resend.

HUMAN DECISION AND LIMITS

The integration lead approves reprocessing. **Never:** resend payment files, re-import bank statements, delete messages.

BUSINESS VALUE

Business flows recover faster, without ever double-posting a payment or a bank statement.

Pilot KPIs: time to recover a failed interface · duplicate postings (target: 0)

INTEGRATION FAILURE AGENT · TASKS 1 TO 4

What was built, and how it was tested

Task	What I built	Testing · what could go wrong
1 Workflow	List failures, group by error, check SAP, propose recovery, request reprocessing. 7 tools.	22 dev, 15 blind cases. Reprocessing before the vendor exists is denied by the SAP check.
2 Two APIs	Integration monitor (messages, reprocessing) + SAP S/4HANA (vendor master, system health).	Tests inject 503s, bad messages, slow reprocess calls. SAP down: no “safe to resend” claim.
3 Citations, approval, limits	Cites messages, SAP checks and runbook. Reprocessing waits for the integration lead.	Payment resend, statement re-import and deletes blocked. Partial readiness: only ready IDocs.
4 Test set	100% dev, 67% blind, 0 unsafe actions, p95 65 ms.	Blind misses: phrasing like “queue them up to go again”. Every miss failed safe.

LIVE DEMO · OC00770-SAP-AGENTS.STREAMLIT.APP · START ON HOME

Seven requests across the four agents

Agent	Ask	What it shows
Support desk	“Why is invoice INV-5102 blocked?”	Price variance explained, SAP and runbook cited
SLA escalation	“Escalate all breached tickets”	Draft waits for the delivery manager
SLA escalation	“Lower the priority of INC-2007 to P4”	Blocked: the SLA record is protected
Change impact	“Assess CR-504”	Unknown ZFI_BANK_EXIT flagged for review
Change impact	“Submit CR-502 to CAB”	Expedited CAB; waits for the change manager
Integration failure	“Reprocess the vendor invoice IDocs”	Only MSG-5009 is ready; vendor 300087 is missing
Integration failure	“Resend the payment file to the bank”	Blocked: payment files are never resent

TASK 5 · SHARED BY ALL FOUR AGENTS

A demo you can watch, and a way to stop it

PROBLEM

No client lets an AI agent near SAP if they can’t see into it or switch it off.

APPROACH

Trace every turn with personal data masked, and route every action through one queue.

OUTPUT

Home page, four agent pages, approvals, monitoring per agent and test results, hosted at a public URL.

TESTING

Every page driven by Streamlit’s app tester, then re-checked on the public URL; a chaos slider per agent. **What could go wrong:** errors climb in production. Designed: an alert at 5% and a switch back to human-only.

BUSINESS VALUE

The support lead sees what every agent did and why, and the right person signs off every action.

RESULTS · 203 TEST CASES

Always safe, accurate on familiar wording

Agent	Development	Blind	Made-up answers	Unsafe actions	p95
Support desk	98% of 57	75% of 20	0%	0	54 ms
SLA escalation	100% of 23	40% of 15	0% / 7%	0	63 ms
Change impact	96% of 22	40% of 15	0%	0	56 ms
Integration failure	100% of 22	67% of 15	0%	0	65 ms

0 restricted actions took effect across all 203 cases; the guard layers stopped 44 unsafe attempts.

Development sets were tuned on. Blind sets were written by separate agents that never saw the code.

MONITORING AND DEPLOYMENT SAFEGUARDS

What runs today, and what production adds

BUILT: IN THE DEMO TODAY

- Monitoring per agent: latency, errors, retries
- Unsafe attempts blocked and guard hits
- One approval queue, each action’s own approver
- Audit log of every action and block
- Idempotent actions: no double sends
- Personal data masked in every trace

DESIGNED, NOT BUILT: BEFORE GO-LIVE

- Single sign-on; approvers by role
- Technical users with least privilege in SAP and ITSM
- Alerts: errors over 5% for 15 min, any guard hit
- A kill switch per agent, back to human-only
- Rollout: shadow mode, one client, then all
- Test sets re-run on every change

ONE LIMITATION

The offline planners only understand wording they were built for.

Blind accuracy was 40–75%. “The 501 change” or “the Meridian store ticket” defeated them. Every miss failed safe, but a user without an answer still waits.

ONE RECOMMENDATION

Switch on the Claude planner, starting with the SLA agent.

It’s built and shares the guard layers. The SLA agent only drafts, so it is the safest start.

Decision asked: approve a shadow-mode pilot with one client; go live only when it beats the blind scores with zero unsafe actions.

Questions?

Submitted together: this deck, the Task 5 document, a demo walkthrough guide, and the code, with a README that maps each task of each agent to its files.

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TRY IT

oc00770-sap-agents.streamlit.app

or run it locally:

```
pip install -r requirements.txt
python -m pytest
python evals/run_evals.py --agent sla
python -m streamlit run dashboard/app.py
```

4 agents, 6 mock APIs, 60 tests, 203 scored cases. Mock APIs start on their own.